

The J_P increment

prime mass after the blind zone, and the honest race

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Abstract

Round 456 retyped the r450–r455 prefix as `VACUOUS_CONFIRMED`: $n < J_P$ is prime-blind Archimedean anatomy. This note accepts that in full and studies the first object that sees primes: the chain increment $E = \text{MAIN} - \text{ARCH}$ from $J_P = \lceil \log 2/\Delta - 1 \rceil$ to the full cap. The Schur increment drifts (no oscillatory rescue). The archimedean baseline is q_* at J_P , not a full-cap ARCH pack (that pack is ill-conditioned). The race $\Delta q_{\text{arith}}/(1 - q_*)$ stays below 1 on living windows and eats the margin at $k = 10$ (kz197). No RH claim. No anti-RH claim.

Status. Ausgang `INCREMENT_DRIFT` / `RACE_EATS_K10` / `LEMMA_REDUCED`. r456 `VACUOUS_CONFIRMED` is accepted. The r450–r455 identities remain true as Archimedean anatomy and are not reused as mincut progress. No L^* claim. No $R^\dagger$ claim. No RH claim. No anti-RH claim.

1 Acceptance

The prefix $n < J_P$ cannot tell MAIN from ARCH ($\Delta q \sim 10^{-16}$). `Iff.rfl` named the full cone. M_∂ is archimedean window mass. Those facts stand. The load-bearing Python object is $q^\dagger$ at `W.cap`.

2 The increment

c^P vanishes for $i < J_P$ (r455). The measure increment is the fold of that delayed comb, via the three-lag identity. E is the Jacobi image of that perturbation — explicit as a map, not a closed formula for $(\Delta\alpha_n, \Delta\beta_n)$.

Lemma 1 (drift, not cancellation). *On kz17, for every grade $n \geq J_P = 19$, $\Delta\rho_n > 0$ (77/77, sum +2.94). Δq jumps +0.05166 at the $n = 2$ tent and +0.046 at the $n = 3$ tent ($n = 30$). (a_n, b_n) oscillate; (ρ, q) drift up. On kz116, $\sum \Delta\rho = +12.5$ (1007/1274 positive): mixed local signs, net positive. `SCRAMBLE` (seed 45701) sends $q(30)$ above 1 while MAIN stays below 1.*

Isolated octave bands do *not* add to Δq (the Jacobi map is nonlinear). The onset is the first prime tent; later primes interact.

Lemma 2 (death is accumulated). *kz197 ($k = 10$, $N = 4071$): $J_P = 407$, $n_{\text{flip}} = 107$, first flip 3788 (r447, dps = 50). The ABD death sits far past J_P . It is an accumulated arithmetic increment, not a local J_P event.*

3 The honest budget

The ARCH *signed* pack at full cap is ill-conditioned ($n_{\text{flip}} = 7\text{--}14$, q explodes on $\text{kHz} \geq 116$). The ARCH μ -chain stays healthy ($\min b \sim 1/2$). The baseline is $q_* := q_{\text{ARCH}}^\dagger(J_P)$.

kHz	a	q_{full}	q_*	Δq	$1 - q_*$	race	note
17	32	0.89136	0.78566	+0.10570	0.21434	0.493	ALIVE
69	256	0.95251	0.89689	+0.05562	0.10311	0.539	ALIVE
116	512	0.96221	0.86848	+0.09374	0.13152	0.713	ALIVE
136	631	0.97209	0.87092	+0.10117	0.12908	0.784	ALIVE
197	1024	0.99958	0.88786	+0.11172	0.11214	0.996	EATS $k = 10$
230	1259	1.14770	0.87326	+0.27445	0.12674	2.165	overshoot

kHz137, 170, 500: full MAIN pack fails or returns NaN after flips — not scored as a race.

Lemma 3 (r453 at cap). *On kHz17, $q^\dagger = S_{N-1}/B_w$ to $2 \cdot 10^{-12}$. Hence $\Delta q_{\text{arith}} = q_{\text{full}} - q_*$ is the accumulated leak of ρ_j for $j \geq J_P$.*

4 Reduction

Lemma 4 (honest smaller statement).

$$q_{\text{full}}^\dagger < 1 \iff \Delta q_{\text{arith}} < 1 - q_*.$$

The increment is supported on $j \geq J_P$ and is net-positive in ρ (no oscillatory rescue). The cofinal hope $\Delta q \leq (1 - q_)(1 - \varepsilon)$ is **REFUTED** by $k = 10$.*

This is r453 evaluated at the cap on the prime tail — a *new use*, not a new identity. It is **not** an alias of r429 ($|Z_{\text{loc}}|$ head bound / triangle **REFUTED**): r429 is a terminal pairing of a selected form; E is the Jacobi image of a lag-sparse second difference of c^P .

5 Kill table

Kill	Status
ARCH world, identical fold (r456)	used throughout
dps at $k = 10$	r447 dps = 50 EXACT_DEAD (flip 3788); float64 re-gated
Determinism	two identical smokes
SCRAMBLE vs oscillation	$q_{\text{SCR}}(30) > 1$; increment is arithmetic
Alias vs r429	different object; documented

Claim boundary. Finite-window increment census. No L^* claim. No $R^\dagger$ claim. No RH claim. No anti-RH claim.